~\.bob\playground\ad_attack_lab\requirements.txt

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# ============================================================
# Active Directory Attack Lab — Python Requirements
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# Python version: >= 3.9
# Install: pip install -r requirements.txt
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# --- Core Kerberos / AD Interaction ---
impacket==0.12.0            # Kerberos AS-REQ/TGS-REQ, SMB, LDAP, DCE/RPC
ldap3==2.9.1                # LDAP/S enumeration of AD objects & SPNs
pyasn1==0.6.0               # ASN.1 parsing (Kerberos ticket structures)
pyasn1-modules==0.4.0       # ASN.1 module definitions

# --- Cryptography ---
cryptography==42.0.8        # AES, RC4, HMAC operations
pycryptodome==3.20.0        # Legacy DES/RC4 cipher support (Kerberos etype 23)

# --- Network & DNS ---
dnspython==2.6.1            # DNS resolution for domain controllers
scapy==2.5.0                # Packet capture / Kerberos traffic analysis

# --- Password / Hash Cracking Interface ---
hashlib-compat==1.0.0       # Consistent hashlib interface across Python versions

# --- LDAP / Active Directory Queries ---
python-ldap==3.4.4          # C-based LDAP library (optional, Windows/Linux)
msldap==0.5.6               # Pure-Python MS LDAP with NTLM/Kerberos auth

# --- GPO / Registry Parsing ---
python-registry==1.4        # Parse Windows registry hives (GPO artifacts)
gposec==0.1.2               # GPO security template parser

# --- Reporting & Output ---
rich==13.7.1                # Rich terminal output with tables and colors
tabulate==0.9.0             # Tabular output formatting
jinja2==3.1.4               # HTML/text report templating
markdown==3.6               # Markdown report rendering

# --- CLI & Utilities ---
click==8.1.7                # CLI framework
colorama==0.4.6             # Cross-platform terminal colors (Windows)
tqdm==4.66.4                # Progress bars for enumeration phases
loguru==0.7.2               # Structured logging

# --- Testing ---
pytest==8.2.2               # Unit test framework
pytest-cov==5.0.0           # Code coverage
responses==0.25.3           # Mock HTTP responses for tests
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50 # --- Dev / Linting (optional) ---
51 black==24.4.2               # Code formatter
52 flake8==7.0.0               # Linter
53 mypy==1.10.0                # Static type checker
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